

Supplemental Material

APPENDIX A LINEAR PROGRAMMING METHOD

Farebrother [34] pointed out that the unconstrained ℓ^1 -norm optimization problem $\min_{\mathbf{r} \in \mathbb{R}^{m \times 1}} \|\mathbf{r}\|_1$ is equivalent to

$$\min_{\mathbf{r}^+, \mathbf{r}^- \in \mathbb{R}^{m \times 1}} \mathbf{1}_m^T (\mathbf{r}^+ + \mathbf{r}^-), \text{ s.t. } \begin{cases} \mathbf{r}^+ - \mathbf{r}^- = \mathbf{r} \\ \mathbf{r}^+, \mathbf{r}^- \succeq 0 \end{cases} \quad (52)$$

Let

$$\mathbf{t} = \mathbf{1}_{2m} = \begin{bmatrix} \mathbf{1}_m \\ \mathbf{1}_m \end{bmatrix}, \quad \beta = \begin{bmatrix} \mathbf{r}^+ \\ \mathbf{r}^- \end{bmatrix}, \quad \Phi = [\mathbf{D}, -\mathbf{D}] \quad (53)$$

and substitute (53) into (52), we immediately have the equivalent description [23]

$$\min_{\beta \in \mathbb{R}^{2m \times 1}} \mathbf{t}^T \beta, \quad \text{s.t. } \begin{cases} \Phi \beta = \mathbf{w} \\ \beta \succeq 0 \end{cases} \quad (\text{LP}_2)$$

where $\mathbf{D}$ and $\mathbf{w}$ are derived from (33) and (34). Suppose the solution to the problem (LP₂) is

$$\beta_{\text{opt}} = [\mathbf{r}_{\text{opt}}^+, \mathbf{r}_{\text{opt}}^-]^T, \quad (54)$$

then the solution to the problem (BP) must be

$$\mathbf{r}_{\text{opt}} = \mathbf{r}_{\text{opt}}^+ - \mathbf{r}_{\text{opt}}^- = \beta_{\text{opt}}(1:m) - \beta_{\text{opt}}(m+1:2m). \quad (55)$$

APPENDIX B GRADIENT PROJECTION METHOD

The first *Gradient Projection* (GP) method is the GPSR method [28]. By segregating the positive coefficients $\mathbf{r}^+$ and the negative coefficients $\mathbf{r}^-$ in $\mathbf{r}$, the equation (QP _{λ}) can be reformulated into the form of standard quadratic programming, namely

$$\min_{\mathbf{z} \in \mathbb{R}^{2m \times 1}} Q(\mathbf{z}) = \gamma^T \mathbf{z} + \frac{1}{2} \mathbf{z}^T \mathbf{B} \mathbf{z}, \text{ s.t. } \mathbf{z} \succeq 0 \quad (56)$$

where

$$\mathbf{z} = \begin{bmatrix} \mathbf{r}^+ \\ \mathbf{r}^- \end{bmatrix}, \gamma = \begin{bmatrix} \lambda - \mathbf{D}^T \mathbf{w} \\ \lambda + \mathbf{D}^T \mathbf{w} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \mathbf{D}^T \mathbf{D} & -\mathbf{D}^T \mathbf{D} \\ -\mathbf{D}^T \mathbf{D} & \mathbf{D}^T \mathbf{D} \end{bmatrix} \quad (57)$$

Note that the matrix $\mathbf{B}$ defined in (57) is symmetric. The gradient of the quadratic form $Q(\mathbf{z})$ is

$$\nabla_{\mathbf{z}} Q(\mathbf{z}) = \gamma + \mathbf{B} \mathbf{z}. \quad (58)$$

Thus the iterative scheme based on the steepest-descent can be designed by (58), viz.

$$\mathbf{z}_{k+1} = \mathbf{z}_k - \alpha_k \nabla Q(\mathbf{z}_k) \quad (59)$$

where the optimal step size α_k can be solved by a standard line-search process such as the Armijo rule [35]. Under appropriate convergence criteria, we can obtain the solution to (56) according to (59) by simple iterations.

The second GP method, known as the TNIPM method [29], can be obtained by transforming the equation (QP _{λ}) into the following constrained quadratic programming problem

$$\begin{aligned} \min_{\mathbf{r} \in \mathbb{R}^{m \times 1}} & \frac{1}{2} \|\mathbf{D} \mathbf{r} - \mathbf{w}\|_2^2 + \lambda \sum_{i=1}^m u_i, \\ \text{s.t. } & -u_i \leq r_i \leq u_i, \quad i = 1, 2, \dots, m. \end{aligned} \quad (60)$$

By constructing the following *logarithmic barrier* function for the bound constrain

$$\Gamma(\mathbf{r}, \mathbf{u}) = -\sum_{i=1}^m \log(u_i + r_i) - \sum_{i=1}^m \log(u_i - r_i) \quad (61)$$

over the domain

$$\Omega = \{(\mathbf{r}, \mathbf{u}) \in \mathbb{R}^m \times \mathbb{R}^m : |r_i| \leq u_i, i = 1, 2, \dots, m\} \quad (62)$$

and substituting (61) into (60), we can find the unique optimal solution $(\mathbf{r}^*(c), \mathbf{u}^*(c))$ to the convex function [29]

$$F_c(\mathbf{r}, \mathbf{u}) = c \cdot \left(\frac{1}{2} \|\mathbf{D} \mathbf{r} - \mathbf{w}\|_2^2 + \lambda \sum_{i=1}^m u_i \right) + \Gamma(\mathbf{r}, \mathbf{u}) \quad (63)$$

where $c \in [0, +\infty)$. The optimal search direction $[\Delta \mathbf{r}, \Delta \mathbf{u}]^T \in \Omega$ for (63) can be determined by

$$\nabla^2 F_c(\mathbf{r}, \mathbf{u}) \cdot \begin{bmatrix} \Delta \mathbf{r} \\ \Delta \mathbf{u} \end{bmatrix} = -\nabla F_c(\mathbf{r}, \mathbf{u}) \in \mathbb{R}^{2m \times 1} \quad (64)$$

with the Newton's method, in which the $\nabla^2 F_c(\mathbf{r}, \mathbf{u})$ in (64) is the Hessian matrix of $F_c(\mathbf{r}, \mathbf{u})$. The optimal step

$$\tau_{\text{opt}} = \arg \min_{\tau \in \mathbb{R}} F_c(\mathbf{r} + \tau \Delta \mathbf{r}, \mathbf{u} + \tau \Delta \mathbf{u}) \quad (65)$$

specified by (65) can be computed with a backtracking line search. In [29], the search direction mentioned in (64) is accelerated by the *preconditioned conjugate gradients* (PCG) algorithm [36] to efficiently approximate the Hessian matrix.

APPENDIX C HOMOTOPY METHOD

The algorithm starts with an initial value $\mathbf{r}^{(0)} = \mathbf{0}$ and operates iteratively, calculating the solutions $\mathbf{r}^{(k)}$ at each step $k = 1, 2, \dots$. Throughout the computation, the active set

$$\mathcal{L} = \{j \in \mathbb{N} : |c_j^{(k)}| = \|\mathbf{c}^{(k)}\|_\infty = \lambda\} \quad (66)$$

remains constant, where $\mathbf{c}^{(k)} = \mathbf{D}^T (\mathbf{w} - \mathbf{D} \mathbf{r}^{(k)})$ is the vector of residual correlations [5]. Let

$$(\cdot)_\lambda = (\cdot)[\mathcal{L}]$$

be the update direction on the sparse support, then $\mathbf{d}_\lambda^{(k)}$ is the solution to the following linear system

$$\mathbf{D}_\lambda^T \mathbf{D}_\lambda \mathbf{d}_\lambda^{(k)} = \text{sign}(\mathbf{c}_\lambda^{(k)}). \quad (67)$$

Thus we can solve (67) to obtain the $\mathbf{d}_\lambda^{(k)} = \mathbf{d}^{(k)}[\mathcal{L}]$, which consists of the non-zero elements in $\mathbf{d}^{(k)}$. Along the direction indicated by $\mathbf{d}^{(k)}$, there are two scenarios when an update on $\mathbf{r}$ may lead to a break-point [32]: inserting an element into the set $\mathcal{L}$ or removing an element from the set $\mathcal{L}$. Let

$$\gamma_+^{(k)} = \min_{i \in \mathcal{L}^c} \left\{ \min \left(\frac{\lambda - c_i^{(k)}}{1 - \mathbf{D}_i^T \mathbf{D}_\lambda \mathbf{d}_\lambda^{(k)}}, \frac{\lambda + c_i^{(k)}}{1 + \mathbf{D}_i^T \mathbf{D}_\lambda \mathbf{d}_\lambda^{(k)}} \right) \right\}_+ \quad (68)$$

and

$$\gamma_-^{(k)} = \min_{i \in \mathcal{L}} \left\{ -r_i^{(k)} / d_i^{(k)} \right\}_+ \quad (69)$$

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where $\min \{\cdot\}_+$ means that the minimum is taken over only positive arguments. The Homotopy algorithm proceeds to the next break-point and updates the sparse support set $\mathcal{L}$ iteratively by the following formula

$$\mathbf{r}^{(k+1)} = \mathbf{r}^{(k)} + \min \left\{ \gamma_+^{(k)}, \gamma_-^{(k)} \right\} \cdot \mathbf{d}^{(k)}, \quad k = 1, 2, \dots \quad (70)$$

The iteration terminates when $\|\mathbf{r}^{(k+1)} - \mathbf{r}^{(k)}\|_2 \leq \epsilon_{\text{re}} \|\mathbf{r}^{(k)}\|_2$ for the given relative error ϵ_{re} according to the Cauchy's convergence criteria.

APPENDIX D ITERATIVE SHRINKAGE-THRESHOLDING METHOD

The IST method [37], [38] are proposed to solve the (QP_λ) as a special case of the following *composite objective function*

$$\mathbf{r}_{\text{opt}} = \arg \min_{\mathbf{r} \in \mathbb{R}^{m \times 1}} f(\mathbf{r}) + \lambda g(\mathbf{r}) \quad (71)$$

where $f(\mathbf{r}) = \frac{1}{2} \|\mathbf{D}\mathbf{r} - \mathbf{w}\|_2^2$ and $g(\mathbf{r}) = \|\mathbf{r}\|_1$. The updating rule of IST to minimize (71) can be calculated by taking a second-order approximation of f . Mathematically, we have [38]

$$\begin{aligned} \mathbf{r}^{(k+1)} &= \arg \min_{\mathbf{r} \in \mathbb{R}^{m \times 1}} \left\{ f(\mathbf{r}^{(k)}) + (\mathbf{r} - \mathbf{r}^{(k)})^\top \cdot \nabla f(\mathbf{r}^{(k)}) \right. \\ &\quad \left. + \frac{1}{2} \|\mathbf{r} - \mathbf{r}^{(k)}\|_2^2 \cdot \nabla^2 f(\mathbf{r}^{(k)}) + \lambda g(\mathbf{r}) \right\} \\ &\approx \arg \min_{\mathbf{r} \in \mathbb{R}^{m \times 1}} \left\{ (\mathbf{r} - \mathbf{r}^{(k)})^\top \cdot \nabla f(\mathbf{r}^{(k)}) \right. \\ &\quad \left. + \frac{\alpha_k}{2} \|\mathbf{r} - \mathbf{r}^{(k)}\|_2^2 + \lambda g(\mathbf{r}) \right\} \\ &= \arg \min_{\mathbf{r} \in \mathbb{R}^{m \times 1}} \left\{ \frac{1}{2} \|\mathbf{r} - \mathbf{u}^{(k)}\|_2^2 + \frac{\lambda}{\alpha_k} g(\mathbf{r}) \right\} \end{aligned} \quad (72)$$

where

$$\mathbf{u}^{(k)} = \mathbf{r}^{(k)} - \frac{1}{\alpha_k} \nabla f(\mathbf{r}^{(k)}), \quad \alpha_k \in \mathbb{R}^+. \quad (73)$$

Note that $g(\mathbf{r})$ is a separable function. As a result, a closed-form solution for $\mathbf{r}^{(k+1)}$ can be obtained for each component

$$\begin{aligned} r_i^{(k+1)} &= \arg \min_{r_i \in \mathbb{R}} \left\{ \frac{(r_i - u_i^{(k)})^2}{2} + \frac{\lambda}{\alpha_k} |r_i| \right\} \\ &= \text{soft} \left(u_i^{(k)}, \frac{\lambda}{\alpha_k} \right) \end{aligned} \quad (74)$$

where

$$\text{soft}(u, a) = \text{sign}(u) \cdot \max \{|u| - a, 0\} \quad (75)$$

is the *soft-thresholding* or *shrinkage* function [39]. The coefficient α_k used in (72), (73) and (74) is adopted to approximate the Hessian matrix $\nabla^2 f$ and it can be calculated with the *Barzilai-Borwein* spectral approach [38].

APPENDIX E ALTERNATING DIRECTION METHOD

With the help of the auxiliary variable $\mathbf{u} \in \mathbb{R}^{(m-n) \times 1}$, the problem (BP_ϵ) can be converted to the following equivalent form [33]

$$\min_{\mathbf{r} \in \mathbb{R}^{m \times 1}, \mathbf{u} \in \mathbb{R}^{(m-n) \times 1}} \|\mathbf{r}\|_1, \quad \text{s.t.} \begin{cases} \mathbf{D}\mathbf{r} - \mathbf{w} = \mathbf{u}, \\ \|\mathbf{u}\|_2 \leq \epsilon. \end{cases} \quad (76)$$

Moreover, we have an augmented Lagrangian subproblem of the form

$$\begin{aligned} \min_{\mathbf{r} \in \mathbb{R}^{m \times 1}, \mathbf{u} \in \mathbb{R}^{(m-n) \times 1}} \quad & \|\mathbf{r}\|_1 - \mathbf{y}^\top (\mathbf{D}\mathbf{r} + \mathbf{u} - \mathbf{w}) + \\ & \frac{\mu}{2} \|\mathbf{D}\mathbf{r} + \mathbf{u} - \mathbf{w}\|_2^2 \end{aligned} \quad (77)$$

s.t. $\|\mathbf{u}\|_2 \leq \epsilon$

where $\mathbf{y} \in \mathbb{R}^{(m-n) \times 1}$ is a multiplier and $\mu > 0$ is a penalty parameter. Applying inexact alternating minimization to (77) yields the following iterative scheme [33]

$$\begin{cases} \mathbf{u}^{(k+1)} = \mathcal{P}_{B_\epsilon} \left(\frac{\mathbf{y}^{(k)}}{\mu} - (\mathbf{D}\mathbf{r}^{(k)} - \mathbf{w}) \right), \\ \mathbf{r}^{(k+1)} = \text{soft} \left(\mathbf{r}^{(k)} - \tau \mathbf{g}^{(k)}, \frac{\tau}{\mu} \right), \\ \mathbf{y}^{(k+1)} = \mathbf{y}^{(k)} - \zeta \mu (\mathbf{D}\mathbf{r}^{(k+1)} + \mathbf{u}^{(k+1)} - \mathbf{w}). \end{cases} \quad (78)$$

where $\tau > 0$ is a proximal parameter, $\zeta > 0$ is a constant and

$$\mathbf{g}^{(k)} = \mathbf{D}^\top \left(\mathbf{D}\mathbf{r}^{(k)} + \mathbf{u}^{(k+1)} - \mathbf{w} - \frac{\mathbf{y}^{(k)}}{\mu} \right). \quad (79)$$

Note that $\mathcal{P}_{B(\epsilon)} : \mathbb{R}^{(m-n) \times 1} \rightarrow B(\epsilon)$ is the projection onto the closed ball $B(\epsilon) = \{\mathbf{d} \in \mathbb{R}^{(m-n) \times 1} : \|\mathbf{d}\|_2 \leq \epsilon\}$ with radius ϵ . It should be pointed out that, when $\epsilon = 0$, (79) reduces to

$$\begin{cases} \mathbf{r}^{(k+1)} = \text{soft} \left(\mathbf{r}^{(k)} - \tau \mathbf{D}^\top \left(\mathbf{D}\mathbf{r}^{(k)} - \mathbf{w} - \frac{\mathbf{y}^{(k)}}{\mu} \right), \frac{\tau}{\mu} \right), \\ \mathbf{y}^{(k+1)} = \mathbf{y}^{(k)} - \zeta \mu (\mathbf{D}\mathbf{r}^{(k+1)} - \mathbf{w}). \end{cases} \quad (80)$$

which induces a simple iterative algorithm for solving the problem (BP) .

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